

Pharmacy Sales & Inventory Analytics System

Strategic Report: Data-Driven Performance Findings & Business Recommendations

Executive Summary

This document presents a comprehensive, data-driven analysis of sales transaction pipelines and inventory health metrics compiled over a 15-month operational period. The analytics engine processed over 50,000 transaction events, representing an active portfolio of 150+ pharmaceutical stock keeping units (SKUs) across 10 distinct categories. Key strategic objectives focus on identifying revenue concentration (Pareto principle), assessing stock-out risks, optimizing working capital tied up in slow-moving inventory, and segmenting the customer base to maximize lifetime value.

Financial Metric	Performance Value
Total Sales Revenue	\$2,480,954.05
Total Gross Profit	\$549,420.81
Average Profit Margin	23.04%
Total Order Count	52,673
Average Order Value (AOV)	\$47.10

Top 10 Strategic Business Findings

- 1. Revenue Concentration:** The top 2 product categories (Analgesics and Cardiology) account for over 45% of total sales revenue, demonstrating high reliance on specific drug lines (Pareto 80/20 Distribution).
- 2. Weekend Dip Pattern:** Transaction volumes drop by 40% on Sundays compared to the Monday and Friday peak periods, indicating potential to reallocate staff or optimize pharmacy operational hours.
- 3. High-Value Customer Value:** The top 10% of customers (VIP Segment) contribute to 38.6% of overall sales profit, showing a high concentration of customer lifetime value (CLV).
- 4. Seasonality Spikes:** Respiratory and Antihistamines categories exhibit sharp revenue peaks during winter (Dec-Jan) and spring (March-April), respectively, driven by seasonal colds and allergy demand.
- 5. Margin Disparity:** The Vitamins category demonstrates the highest average profit margin (48%), while Cardiology exhibits high transaction volumes but lower margins (21%), outlining a volume vs. margin trade-off.
- 6. Stockout Alert Risk:** Approximately 12.5% of total SKUs are currently operating below critical reorder levels, creating immediate risks of lost sales on essential medications.

- **7. Capital Locked in Slow-Moving Inventory:** Over \$42,000 in working capital is currently locked up in products categorized as Class C (slow-moving), showing low demand and excess safety stock.
- **8. Customer Segment Recency Churn:** Over 15% of the customer base exhibits 'At-Risk' behavior with no transaction history in the past 120 days, signaling a need for active re-engagement.
- **9. Store Location Variations:** The Downtown store outpaced the Eastside location by 32% in total revenue, displaying localized demographic demand patterns.
- **10. Dynamic Discount Leakage:** Bulk quantity purchase discounts (5-10% discount) implemented in the POS led to an estimated 4% erosion in profit margins without a matching volume offset.

CRITICAL INSIGHT: Immediate operational intervention is required on critical reorder alerts for top-selling products. Stockouts in Analgesics could result in an estimated 15% revenue loss next month.

Top 10 Actionable Business Recommendations

- **1. Optimize Inventory for High-Turnover SKUs:** Increase reorder points by 20% for top Analgesics and Antibiotics during winter seasons to avoid stock-out losses.
- **2. Re-allocate Store Staffing Hours:** Reduce Sunday operational hours by 3 hours and re-allocate budget to peak sales windows on Mondays and Fridays.
- **3. Implement VIP Loyalty Program:** Launch a targeted patient relationship program for high-CLV customers, offering free home delivery and automatic chronic prescription refills.
- **4. Phase Out and Liquidate Slow-Moving Class C SKUs:** Bundled discounts (e.g. 'buy vitamins, get free hygiene wipes') or direct supplier return agreements should be utilized for the bottom 30 slow-moving items.
- **5. Expand High-Margin Vitamin Offerings:** Increase promotional shelf space for wellness and vitamins to drive aggregate store margin upward.
- **6. Automate Reorder Alerts:** Set up database triggers to send direct purchasing notifications to suppliers once stock drops below safety levels.
- **7. Execute Re-Engagement Campaigns for At-Risk Customers:** Dispatch promotional email vouchers to 'At-Risk' segment customers offering a 15% discount on their next wellness purchase.
- **8. Localize Store Merchandising:** Align the inventory mix of the Eastside store with its specific demand patterns (e.g. stock more pediatric vitamins and less chronic cardiology meds).
- **9. Establish Dynamic POS Pricing Rules:** Restrict POS price discounts on high-demand, low-margin products, limiting discounts only to items near expiry.
- **10. Standardize Supplier Lead-Time Agreements:** Negotiate guaranteed 48-hour delivery SLA with primary distributors for critical items to allow reduction of safety stock buffers.